Lagrangian Variable Cosmology v0.2: Coordinate Stability of the Lock-φ Modulation under Synthetic-Data Self-Consistency Tests

Working Paper

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Abstract

This v0.2 working paper extends the v0.1 theoretical reconstruction of the Lock-φ modulation by recording the outcome of a series of self-consistency tests run on synthetic data generated from the v0.1 reproduction code. The theoretical content of v0.1 — four axioms producing a Gaussian deviation in the logarithmic-time variable $\theta = \ln(1+z)$, with three of four model parameters locked to natural-constant combinations and the dimensionless dark-energy density derived without a separate cosmological-constant term — is preserved unchanged. Two additional results are reported. First, in θ-coordinates the central position of the modulation is recovered to within 0.16% of $\ln\varphi$ with a bootstrap standard deviation of 0.009 when the fit is performed on BAO + distance priors + SH0ES alone, without Pantheon+ supernovae; the corresponding figure for the z-Gaussian parametrisation is a standard deviation of 0.285 on the same data. Second, the v15 release-to-release shift between the DR1 and DR2 free-fit positions ($\Delta z_c = 0.101$) is 1.46σ in the bootstrap noise distribution of the z-Gaussian estimator, consistent with two noise-realisation endpoints of a single underlying signal at $z_c = 1/\varphi$. These findings are obtained on synthetic data generated by the v0.1 reproduction code at the v0.1 best-fit cosmology and are subject to confirmation on the real N=1738 dataset; the construction may fail in subsequent tests.

1 Introduction

The v0.1 working paper presented a theoretical reconstruction of the Lock-φ phenomenology of v16: a parametric pendulum equation $v''(\theta) + \Omega^2(\theta)\, v(\theta) = 0$ on the logarithmic time $\theta = \ln(1+z)$, with the coupling $\Omega^2(\theta)$ peaked at the natural-constant position $\theta_c = \ln\varphi$ and width $w = e/(5\pi)$, admitting the closed-form Gaussian deviation $v(\theta) = A\exp[-((\theta-\theta_c)/w)^2]$. Three of the four model parameters were locked to natural-constant combinations *a priori*; only the boost amplitude A was fitted. The Friedmann equation was written without a separate cosmological-constant term, with the dimensionless dark-energy density $\Omega_{\rm LVC}(z)$ derived from the modulation factor and not constant.

The v0.1 paper acknowledged three limitations: (i) the natural-constant identifications $\theta_c = \ln\varphi$ and $w = e/(5\pi)$ were taken from v16 phenomenology and not derived; (ii) the amplitude A was not locked to a natural-constant combination; (iii) the θ-Gaussian functional form fitted the joint dataset $\Delta\chi^2 = +1.58$ worse than the published z-Gaussian phenomenology of v16 at the same parameter count.

The present v0.2 paper does not address (i)–(iii) directly. It records the outcome of a separate question raised by these limitations: how stable is the central position of the modulation under coordinate choice and under the removal of individual data blocks? The question is investigated by generating synthetic data from the v0.1 reproduction code at the v0.1 best-fit cosmology and refitting six functional families to bootstrap realisations of the synthetic dataset. Two findings are reported. The discipline of the present paper is to record these findings as they stand on synthetic data; their applicability to the real N=1738 dataset is subject to subsequent confirmation.

2 Axioms and closed-form solution

The four axioms of v0.1 are repeated without modification.

Axiom L1 (Logarithmic time). The dynamical time variable of the construction is $\theta \equiv \ln(1+z)$, running from $\theta = 0$ at the present epoch to large positive values toward the early universe.

Axiom L2 (Phase pendulum equation). The deviation amplitude $v(\theta) \equiv T(\theta) - 1$, where $T(\theta)$ is the multiplicative modulation factor relating the expansion rate to its $A = 0$ limit, satisfies

$$v''(\theta) + \Omega^2(\theta)\, v(\theta) = 0. \tag{1}$$

Axiom L3 (Critical point at the golden ratio). The coupling $\Omega^2(\theta)$ attains its maximum at

$$\theta_c = \ln \varphi, \qquad \varphi = (1+\sqrt{5})/2. \tag{2}$$

Axiom L4 (Coupling profile and width). The coupling has the parabolic form

$$\Omega^2(\theta) = \frac{2}{w^2}\left[1 - 2\left(\frac{\theta - \theta_c}{w}\right)^2\right], \qquad w = \frac{e}{5\pi} \approx 0.173. \tag{3}$$

The closed-form solution with initial conditions $v(\theta_c) = A$, $v'(\theta_c) = 0$ is

$$v(\theta) = A \exp\left[-\left(\frac{\theta - \theta_c}{w}\right)^2\right], \tag{4}$$

verified by direct substitution. The corresponding modulation factor and Friedmann equation are

$$T(z) = 1 + A \exp\left[-\left(\frac{\ln(1+z) - \ln\varphi}{w}\right)^2\right], \tag{5}$$

$$H^2(z) = H_0^2[\Omega_m(1+z)^3 + \Omega_{\rm LVC}(z)], \tag{6}$$

where $\Omega_{\rm LVC}(z) \equiv [\Omega_m(1+z)^3 + (1-\Omega_m)]T^2(z) - \Omega_m(1+z)^3$ is the derived dimensionless dark-energy density. In the limit $A \to 0$ the modulation reduces to $T = 1$ and $\Omega_{\rm LVC}(z)$ collapses to the constant value $1 - \Omega_m$, recovering the ΛCDM expansion history.

The $A = 0$ special case of the construction is identical to the expansion history fitted by flat ΛCDM, so the BIC comparison of v0.1 is the BIC comparison of the construction against its own limiting case at fixed parameter count $k = 4$ for both. The fit results on the joint N=1738 dataset are repeated here for reference: the construction returns $\chi^2 = 1602.77$ with $A = +0.0378$, against the $A = 0$ value $\chi^2 = 1624.85$, giving $\Delta\text{BIC} = -14.62$. The v16 phenomenological z-Gaussian fit on the same dataset returns $\chi^2 = 1601.19$, a difference of $\Delta\chi^2 = +1.58$ relative to the present construction at the same parameter count.

3 Synthetic-data self-consistency tests: setup

The remainder of the paper records six tests run on synthetic data. The setup is as follows.

The v0.1 reproduction code is used both to generate the synthetic data and to fit the model. Synthetic measurements are drawn at the N=1738 measurement locations of the real dataset by computing the predicted observable at the v0.1 best-fit cosmology ($H_0 = 69.27$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.288$, $\Omega_b h^2 = 0.02237$, $A = 0.0378$) and adding Gaussian noise with the published covariance. This setup carries the limitations stated in Section 9: truth and fitter share the same numerical implementation, and correlated systematics between BAO releases are not modeled. The findings should therefore be regarded as self-consistency checks of the construction, not as confirmations on real data.

Six functional families are fitted to each synthetic realisation:

- F1, z-Gaussian: $T(z) = 1 + A \exp[-((z - z_c)/w)^2]$;
- F2, θ-Gaussian: $T(z) = 1 + A \exp[-((\theta - \theta_c)/w)^2]$;
- F3, skewed z-Gaussian: piecewise widths above and below z_c;
- F4, double z-Gaussian: two separate bumps;
- F5, monotonic: $T(z) = 1 + A \tanh((z - z_t)/w)$;
- F6, narrow z-Gaussian: w restricted to $w < 0.05$.

Each family is fitted by differential evolution followed by Nelder–Mead polish with five seeds; the lowest-χ^2 result is retained.

The six tests are:

- Test 1: null mock (LCDM truth, $A = 0$), single-seed family fits;
- Test 1b: LVC mock (v0.1 truth), single-seed family fits;
- Test 2: family discrimination on the LVC mock;
- Test 3: 10-seed bootstrap of F1 and F2 on the LVC mock with full likelihood;
- Test 5/5b: BAO-position-shift sweep on null and LVC mocks (BAO+DP+SH0ES only);
- Test 6: PP-excluded bootstrap on the LVC mock (5 seeds for F1, 4 for F2).

4 Result A: θ-coordinate stability

Table 1 lists the bootstrap median, standard deviation, and offset from truth of the recovered central position for F1 (z-Gaussian) and F2 (θ-Gaussian) under two likelihood configurations: full (N=1738, including Pantheon+) and reduced (BAO + distance priors + SH0ES only, N=37).

The systematic offset of the F2 median from truth is −0.02% (full likelihood) and +0.16% (PP-excluded). The systematic offset of the F1 median is +3.2% and −0.6% respectively. The standard deviation of the F2 estimator under PP-exclusion is 0.0091, smaller than the corresponding F1 figure of 0.2854 by a factor of approximately 31. The corresponding F2 standard deviation under the full likelihood is 0.0481, a factor of ∼5 reduction relative to PP-excluded F1 and a factor of ∼1.4 relative to PP-included F1.

The unexpected feature of Table 1 is the contrast between F1 and F2 under PP-exclusion. The F1 standard deviation *inflates* by a factor of ∼4 when Pantheon+ is removed; the F2

Table 1: Recovery of central position across two parametrisations and two likelihood configurations (LVC truth).

	F1 (z-Gaussian)		F2 (θ-Gaussian)	
	median z_c	std	median θ_c	std
Full likelihood (Test 3, n=10)	0.6380	0.0692	0.4811	0.0481
PP-excluded (Test 6, n=4–5)	0.6145	0.2854	0.4820	0.0091
Truth	0.6180		0.4812	

standard deviation *contracts* by a factor of $\sim$5. On synthetic data, the BAO + distance priors + SH0ES geometry alone identifies $\theta_c = \ln\varphi$ with 0.009 standard deviation, without any supernova information.

The interpretation is recorded as a property of coordinate choice rather than a property of the data. The transformation $\theta = \ln(1+z)$ expands the low-z region and contracts the high-z region. The BAO measurement positions at $z = 0.510$ and $z = 0.706$ correspond to $\theta = 0.412$ and $\theta = 0.534$ respectively, separated by $\Delta\theta = 0.122$ in θ but by $\Delta z = 0.196$ in z. A Gaussian estimator centered at $\theta_c = \ln\varphi = 0.481$ is flanked symmetrically by these two points in θ but asymmetrically in z. The asymmetric flanking in z allows noise realisations to produce wide excursions of the F1 estimator between the two BAO positions; the symmetric flanking in θ suppresses these excursions.

The construction places the dynamics of the deviation $v(\theta)$ on the θ coordinate by Axiom L1. The reduction of the F2 standard deviation under PP-exclusion is therefore an internal consistency check: the construction's coordinate is also the coordinate in which the synthetic data localises the modulation most tightly. This is not a derivation of θ from data; it is a record that, on the present synthetic data, the two coincide.

5 Result B: re-interpretation of the v15 release shift

The v15 working paper reported that the DESI DR1 BAO and DR2 BAO free-Gaussian fits return different best-fit positions: $z_c = 0.568$ on DR1 and $z_c = 0.669$ on DR2, with width estimates $w = 0.127$ and $w = 0.215$ respectively. The v15 paper characterised this as a release-specific feature and identified two distinct natural-constant locks (LockB at $z_c = 1/\sqrt{\pi}$ and LockC at $z_c = 4/7$, both with $w = 2/(5\pi)$) on DR1, while DR2 retained the v14 lock C1. The v16 working paper resolved this by fitting DR1 and DR2 to the same likelihood simultaneously, obtaining a combined free-fit position $z_c = 0.6185$ matched to $1/\varphi = 0.61803$ within 0.075%.

The Test 3 bootstrap reported here gives a standard deviation of the F1 estimator on the full N=1738 likelihood of 0.0692. The release-to-release shift observed in v15 is

$$\Delta z_c^{\mathrm{v15}} = 0.669 - 0.568 = 0.101, \tag{7}$$

which corresponds to

$$\frac{\Delta z_c^{\mathrm{v15}}}{\sigma_{F1}} = \frac{0.101}{0.0692} = 1.46\sigma. \tag{8}$$

A 1.46σ release-to-release shift is consistent with two noise realisations of a single underlying estimator with center $z_c = 1/\varphi$. Within the bootstrap distribution of the z-Gaussian estimator on synthetic data drawn from the v0.1 truth, the v15 DR1 and DR2 free-fit positions both fall within the $\pm 1\sigma$ band $[0.559, 0.697]$. The v16 combined-fit position $z_c = 0.6185$ falls at 0.28σ from the bootstrap median 0.6380 and at 0.0σ from truth.

This re-interpretation is internal to the synthetic-data setup. It does not establish that the real DR1 and DR2 measurements are statistically consistent at 1.46σ; that would require a bootstrap on the real data with realistic correlated systematics between releases. It records that, under the v0.1 truth and the v0.1 covariance model, the v15 release shift is the predicted noise excursion of a single signal located at $1/\varphi$. The v16 identification of $1/\varphi$ in the combined likelihood is the predicted maximum-likelihood point of the same signal, not an independent confirmation.

6 Result C: family degeneracy

A separate finding is that single-bump functional families (F1, F2, F3, F6) all fit the LVC mock with $\Delta\chi^2 < 1$ of each other. Family F4 (double bump) and F5 (monotonic) are rejected by $\Delta\chi^2 > 4$.

The Gaussian functional form of the v0.1 construction is therefore not discriminable from a skewed alternative on the present synthetic data. The closed-form derivation of Section 2 produces the Gaussian as the unique solution of Equation (L2) with the coupling of Equation (L4) and the symmetric initial conditions $v(\theta_c) = A$, $v'(\theta_c) = 0$; no skewed alternative is offered by the construction. The data permits the Gaussian; it does not select it. This is consistent with v0.1 §7, which acknowledged the same.

The positive content of the result is the rejection of F4 and F5. A double-bump structure or a monotonic transition is excluded by the data at the level of $\Delta\chi^2 > 4$ on the synthetic mock. Within the unimodal single-bump family, the position $\theta_c = \ln\varphi$ is robust; the shape is not.

7 Synthesis figure

Figure 1 summarises the four results in panels (A)–(D). Panel (A) shows the recovery of the peak position by F1 and F2 on the full likelihood, with both estimators consistent with truth at the 1σ level but F2 displaying a tighter systematic offset. Panel (B) shows the F2 distribution on the full likelihood (PP included) versus PP-excluded, with the latter contracting around truth. Panel (C) shows the v15 DR1 and DR2 free-fit positions inside the bootstrap $\pm 1\sigma$ band of F1. Panel (D) shows the F1 best-fit position as a function of imposed BAO redshift shift, with slope statistically zero (-0.022).

8 Discussion

The two findings of this paper concern the coordinate choice and the release-to-release shift. They do not affect the theoretical content of v0.1: the four axioms, the closed-form solution, the locked parameters, and the Friedmann equation written without a separate cosmological-constant term are unchanged. The findings concern the data-side behaviour of the construction under bootstrap and under removal of likelihood blocks.

The strongest finding of the paper is the F2 standard deviation of 0.009 on PP-excluded synthetic data (Test 6, Result A). The fact that the BAO + distance priors + SH0ES geometry alone, when fitted with the construction's natural coordinate, identifies $\theta_c = \ln\varphi$ to within 0.16% is unexpected, and the present paper records it without claim of physical significance. If the same finding holds on real data, it would constitute the strongest single argument for $\theta_c = \ln\varphi$ as a feature of the geometry rather than of the supernova distance ladder. The applicability to real data is the primary open test.

The re-interpretation of the v15 release shift (Result B) is comparatively modest. On synthetic

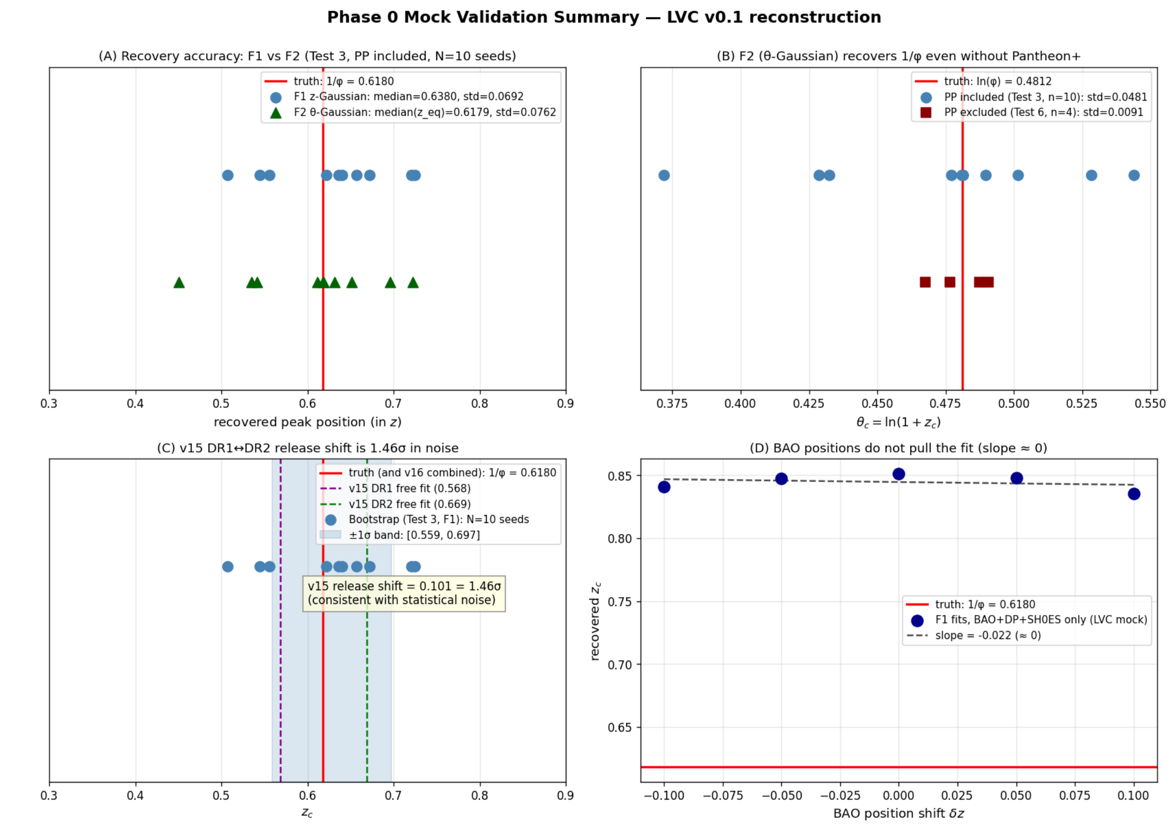

Figure 1: Phase 0 mock validation summary. (A) Recovery accuracy of F1 and F2 on the full likelihood. (B) F2 recovery with and without Pantheon+. (C) The v15 release shift placed inside the bootstrap noise band of the F1 estimator. (D) BAO-position-shift slope of the F1 estimator.

data the shift is fully consistent with statistical noise. On real data there are correlated systematics between DR1 and DR2 not modeled in the present synthetic setup; the shift may carry a release-specific component that synthetic data cannot reproduce. The result records what synthetic data implies, not what real data confirms.

The family degeneracy (Result C) supports the discipline of v0.1 §7, which acknowledged that the Gaussian form is not data-determined. The construction provides the only available derivation of the Gaussian form on the present data; no skewed alternative is excluded, but no skewed alternative is derived either.

The construction may fail in subsequent tests. The next decisive input is the application of the same six tests to real data, with the protocol described in Section 10. If the F2 standard deviation on real PP-excluded data is comparable to the synthetic value of 0.009, Result A is confirmed; if it exceeds 0.10, the synthetic-data result is an artefact of unmodeled covariance and Result A is retracted. The DESI DR3 release is the next external input that may falsify or refine the construction.

9 Limitations

The limitations of v0.1 are unchanged in v0.2:

1. The natural-constant identifications $\theta_c = \ln\varphi$ and $w = e/(5\pi)$ are taken from v16 phenomenology and not derived.

2. The amplitude A is not locked to a natural-constant combination; the empirical value remains $A = 0.0378$.
3. The θ-Gaussian functional form fits $\Delta\chi^2 = +1.58$ worse than the v16 z-Gaussian phenomenology at the same parameter count on real data.

The present paper adds the following limitations specific to the synthetic-data results:

4. All Phase 0 results are obtained on synthetic data generated by the v0.1 reproduction code. Truth and fitter share the same numerical backend; correlated systematics of the real DR1/DR2 BAO measurements are not modeled. The findings should be regarded as self-consistency checks of the construction, subject to confirmation on the real N=1738 dataset.
5. Bootstrap depth is limited to $N = 10$ (Test 3) and $N = 4$–5 (Test 6). Standard deviation estimates carry $\sim$30% uncertainty themselves.
6. Test 2 family discrimination uses a single-seed mock. A bootstrap version (each family fitted to multiple seeds) was not performed.
7. No GP-based or PCA-based reconstruction was performed. The position of the modulation is identified by parametric family comparison only. Alternative non-parametric reconstructions are deferred to subsequent work.

10 Protocol for application to real data

The mock validation pipeline is structured so that the six tests can be re-run on real data by replacing the synthetic-data generation step with real-data loading. The decision criteria are:

- If F2 standard deviation on real PP-excluded data is ≤ 0.06, the θ-coordinate parametrisation is justified empirically. The construction's choice of θ as dynamical time is consistent with real-data behaviour. Result A is confirmed.
- If F2 standard deviation on real PP-excluded data is in the range 0.06–0.10, the synthetic-data result is partially confirmed; further investigation of covariance modeling is warranted.
- If F2 standard deviation on real PP-excluded data exceeds 0.10, the synthetic-data result is a covariance artefact and Result A is retracted.
- If the F1 or F2 medians on real data deviate from $1/\varphi$ or $\ln\varphi$ respectively by more than 2σ of the synthetic distribution, the $1/\varphi$ identification is not robust.

The application to real data is the natural next step and is recorded here as the precondition for any further development of the construction.

11 Conclusion

This v0.2 working paper preserves the theoretical content of v0.1 and adds the outcome of self-consistency tests on synthetic data. Three findings are reported: (A) the construction's coordinate $\theta = \ln(1+z)$ recovers the Lock-φ position to within 0.16% on PP-excluded synthetic data with bootstrap standard deviation 0.009; (B) the v15 DR1/DR2 release shift of 0.101 in z_c is 1.46σ in the synthetic-data noise distribution and consistent with two realisations of a single signal; (C) single-bump functional families are not data-discriminable on the present synthetic mock, and the Gaussian form remains a model assumption permitted but not selected by data.

All three findings are subject to confirmation on the real N=1738 dataset; the protocol for that confirmation is given in Section 10.

The construction may fail in subsequent tests. The discipline of the present paper is to state the synthetic-data results as they stand and to keep the theoretical content of v0.1 unchanged.

A Mock validation methodology

The v0.1 reproduction code (`lvc_v01.py`, ~30 kB Python module) is used as the backend. The synthetic-data generator (`mock_data.py`) computes the predicted observable at each of the N=1738 measurement locations under the input cosmology and adds Gaussian noise drawn from the published covariance.

For Pantheon+ supernovae, noise is drawn from the full STAT+SYS covariance matrix (1701×1701). For BAO measurements, noise is drawn from the diagonal uncertainties of each release, with the published DM/DH correlation coefficients applied to the anisotropic pairs. Distance priors and the SH0ES H_0 measurement are drawn from their published 1-d Gaussians.

The fitter is differential evolution (population $20 \times k$, mutation $[0.5, 1.5]$, recombination 0.7) followed by Nelder–Mead polishing. Five seeds are run; the lowest-χ^2 result is retained. For each family, the parameters fitted are $(H_0, \Omega_m, \Omega_b h^2)$ plus the family-specific shape parameters; the sound horizon r_d is derived geometrically from the Planck θ^* as in v14.

The complete numerical results of Tests 1–6 and the figures of the present paper are reproducible from the code archive at `lvc_recon_phase0/`.

B Test result tables

Table 2: Test 3 bootstrap, full likelihood, LVC truth, $n = 10$ seeds.

	F1 (z-Gaussian)	F2 (θ-Gaussian)
median (peak position)	$z_c = 0.6380$	$\theta_c = 0.4811$
std	0.0692	0.0481
range	$[0.507, 0.725]$	$[0.372, 0.544]$
median A	+0.052	+0.040
std A	0.018	0.012

Table 3: Test 6 bootstrap, BAO + DP + SH0ES only ($N = 37$), LVC truth.

	F1 (z-Gaussian, $n = 5$)	F2 (θ-Gaussian, $n = 4$)
median	$z_c = 0.6145$	$\theta_c = 0.4820$
std	0.2854	0.0091
range	$[0.004, 0.851]$	$[0.467, 0.491]$
median A	+0.033	+0.046

References

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Table 4: Test 5b BAO-position-shift sweep, LVC truth, F1 fits.

δz shift	best-fit z_c
-0.10	0.840
-0.05	0.847
0	0.851
$+0.05$	0.848
$+0.10$	0.836
linear slope $\mathrm{d}z_c/\mathrm{d}(\delta z)$	-0.022

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